INTRODUCTION TO ECONOMICS
MGT404

Market Equilibrium

Yale

Recap and Today’s Plan

Last Lecture

- Reservation prices: the *most* you’re willing to pay
- Demand functions: represent distributions of reservation prices in markets.
- Aggregating demand: add by quantity.

This Lecture

- Firm supply curve, market supply curve
- Putting Supply and Demand together: Market Equilibrium

Quant skills: find equilibrium price and quantity, calculate producer and consumer surplus

Today's Plan

Supply and Equilibrium:

1. Supply curves

- **At what price is an individual or firm willing to supply a good or service?**

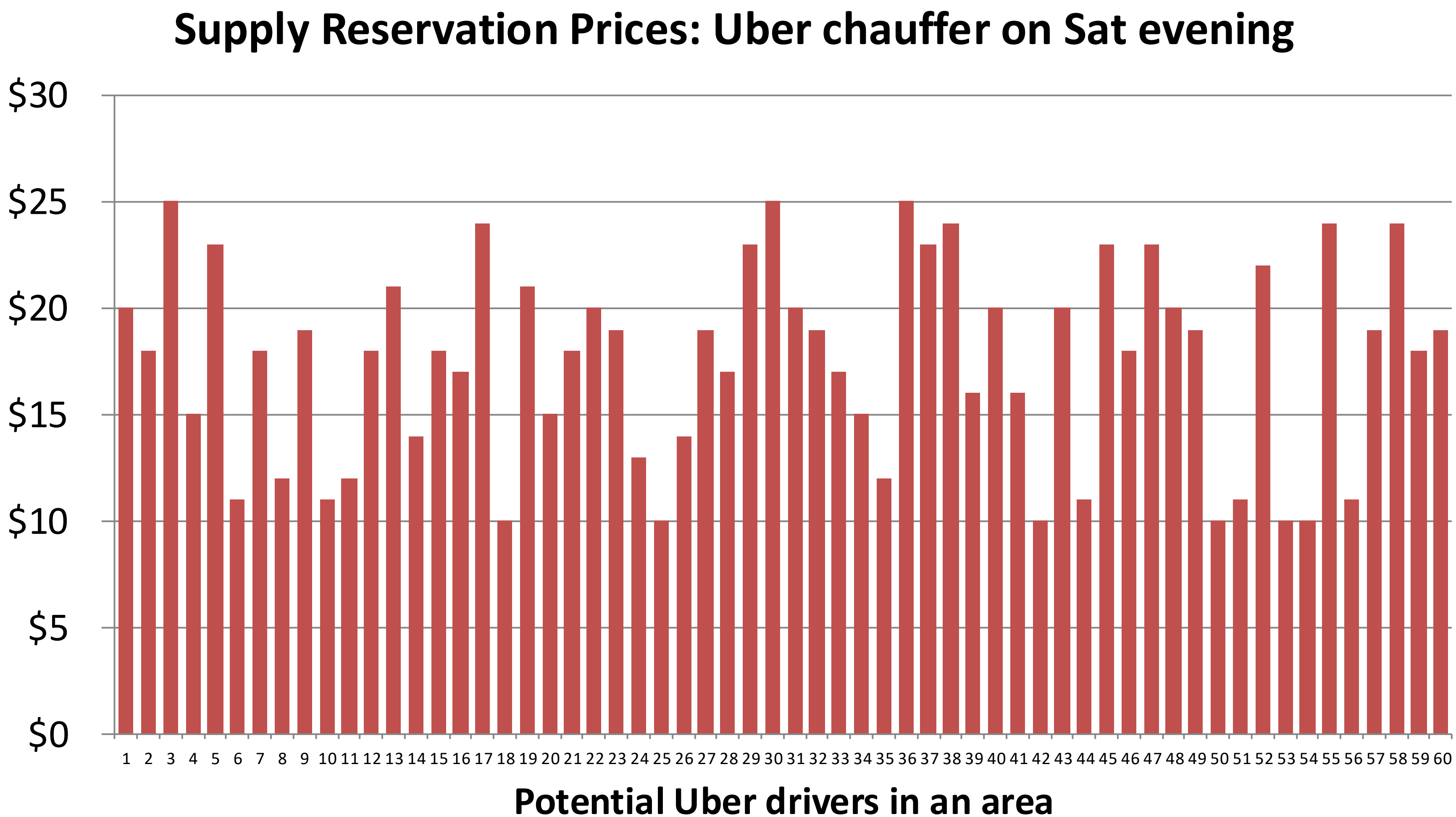
2. Equilibrium

- Equilibrium: when demand=supply
- Consumer surplus and producer surplus

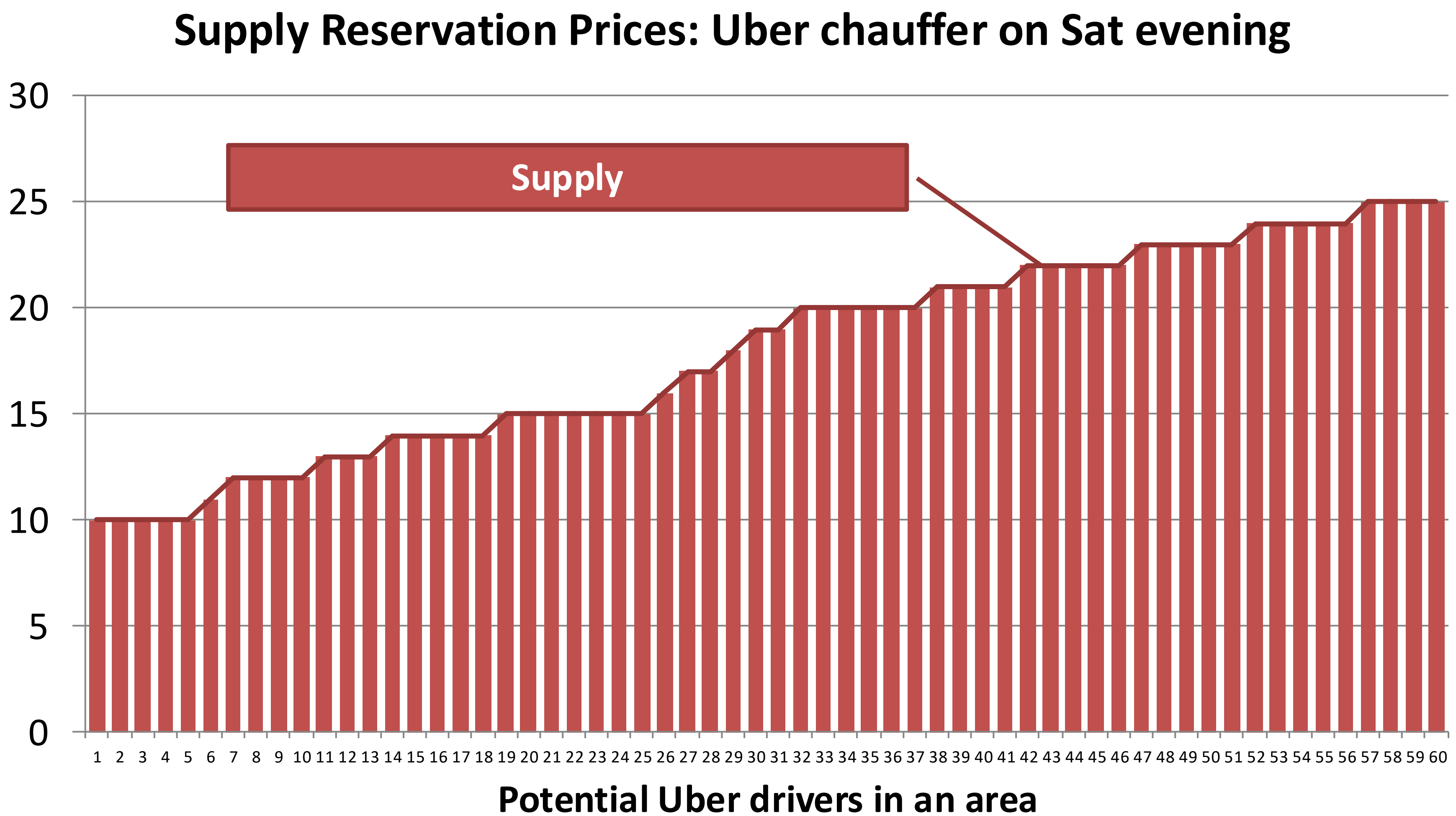
3. Application to Uber

- Uber surge pricing discussion
- Finding the equilibrium price and quantity

Supply from Individual Reservation Prices



Supply from Individual Reservation Prices



Aside: Labor Shortage?

REUTERS®

World Business Legal Markets Breakingviews Technolc

May 4, 2021
6:27 AM CDT
Last Updated 3 months ago

Technology

Driver shortage, looming regulation cloud Uber, Lyft recovery

The teacher shortage is real, large and growing, and worse than we thought

Amid a raging pandemic, the US faces a nursing shortage. Can we close the gap?

November 20, 2020 8.23am EST

There is a labor shortage ***at current wages***

Today's Plan

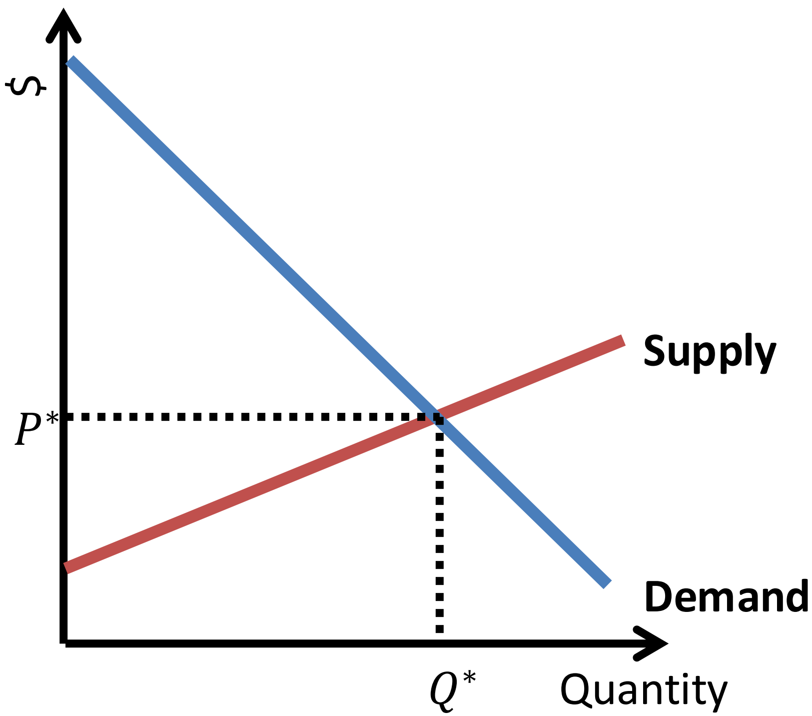
Equilibrium and elasticity:

1. Supply curves
   - At what price is an individual or firm willing to supply a good or service?
2. Equilibrium
   - **Equilibrium: when demand=supply**
   - **Consumer surplus and producer surplus**
3. Application to Uber
   - Uber surge pricing discussion
   - Finding the equilibrium price and quantity

Market Equilibrium

Market Equilibrium

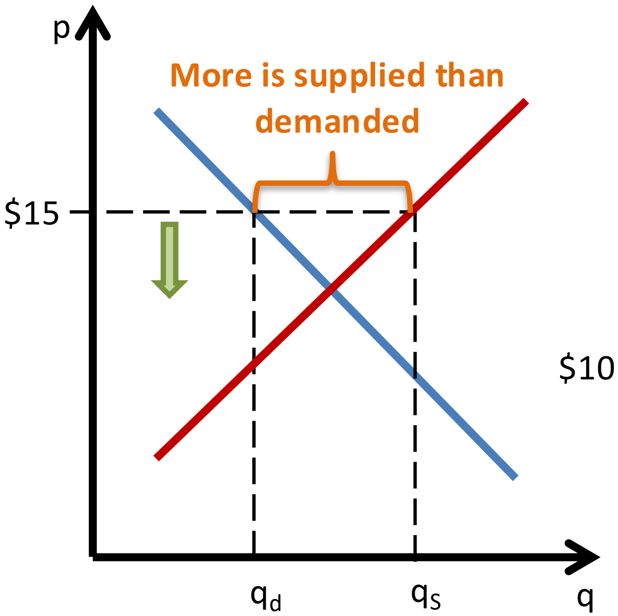
We have shown how reservation prices lead to a **demand curve**, and individual reservation prices (soon: firm production) lead to a **supply curve**.
The price and quantity at which these two intersect are the **market equilibrium**.



- At the equilibrium, no one would want to supply more or less, and consumers would not want to buy more or less.
- An equilibrium has a Price, P*, and Quantity, Q*.
- At P*, consumers want to buy exactly Q* AND suppliers want to provide exactly Q*.
- To solve for the equilibrium P and Q using equations, you'll set the demand and supply equations equal, and solve for Q*, then plug in to find P*.

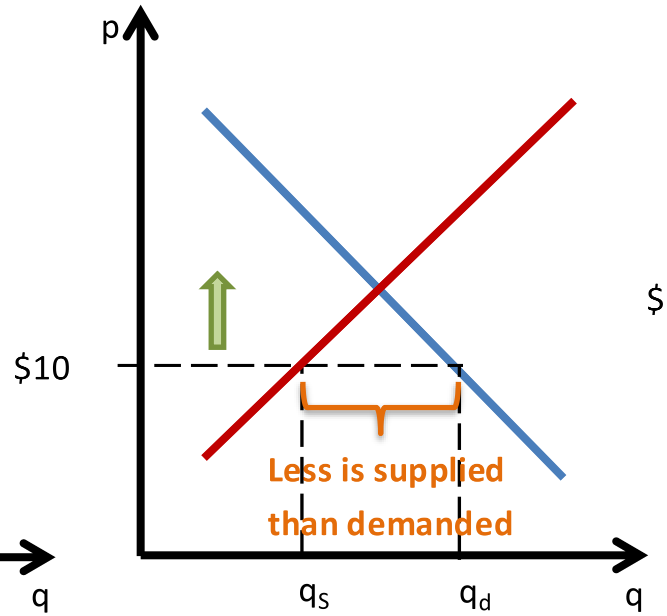
Why equilibrium?

Price too high



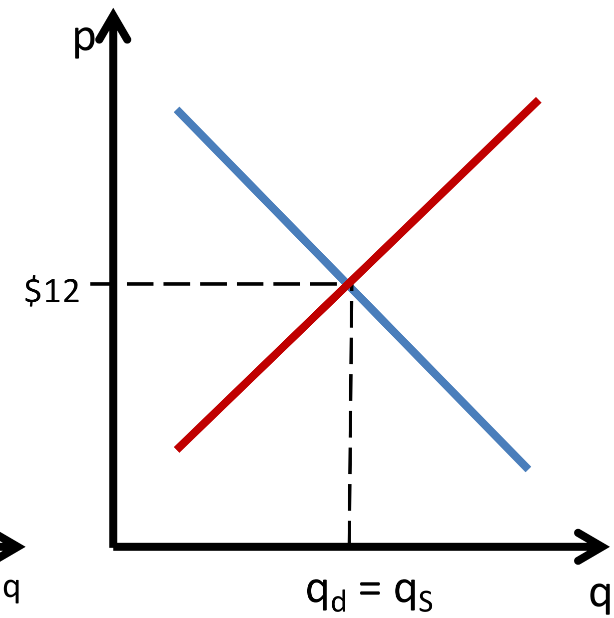
- At p = \$15, suppliers want to sell more than consumers will buy
- If p ↓, fewer suppliers will sell and more consumers will buy

Price too low



- At p = \$10, consumers want to buy more than suppliers will sell
- If p↑, more suppliers will sell and fewer consumers will buy

Price just right



- At p = \$12, the quantity demanded equals the quantity supplied
- No clear force pushing the price

Equilibrium is where demand=supply. Markets out of equilibrium move in the direction of equilibrium

Assumptions behind this Theory

Assumption	Description	Implication
Undifferentiated products	• Consumers perceive each firm's product to be the same	**Law of One Price**: there is a single price at which transactions will occur
Perfect information	• Consumers know the prices being charged by all sellers	
Buyers are small	• A buyer cannot affect the market price	**Price Takers**: buyers and sellers take prices as given when making purchase and production decisions
Sellers are small	• A seller cannot affect the market price, nor input prices	

- Will maintain these assumptions from now on until Lecture 7 (perfect competition).
- Will then relax each of these assumptions from Lecture 8 (monopoly) onward.

Market Equilibrium Game: MobLab

Instructions

You will be a buyer or a seller in an orange market. Buyers have different values for oranges. They make money by buying for less than the value.

Sellers have different costs for oranges. They make money by selling for more than the cost.

Use the slider to submit a bid to buy or an offer to sell. You can also use the blue button to buy or sell an orange at the going price.

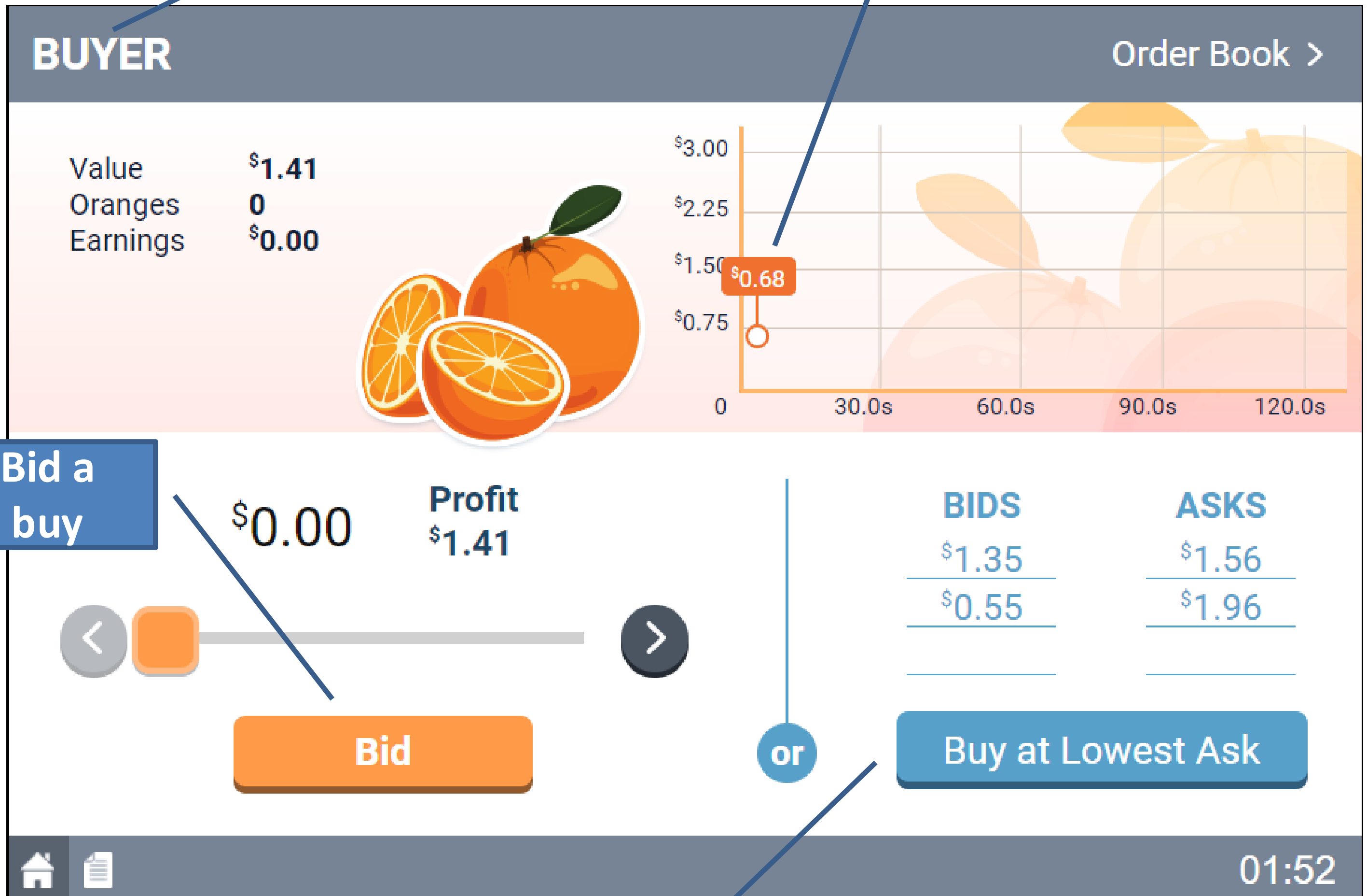
The market will close when the time runs out or when all oranges have been transacted. Let's trade!

Your objective:
Buyer: maximize CONSUMER SURPLUS (your value – price paid)
Seller: maximize PROFITS (price received – your cost)

Sample Screen

Your Role

Transactions



You can Bid a price to buy

… or buy at lowest "ask"

Sellers see the reverse ("ask", "sell at highest bid") md2cknp24

MobLab Discussion

Discussion points:

1. How did you do? Did you have a strategy?

2. Was the information provided (transactions) useful?

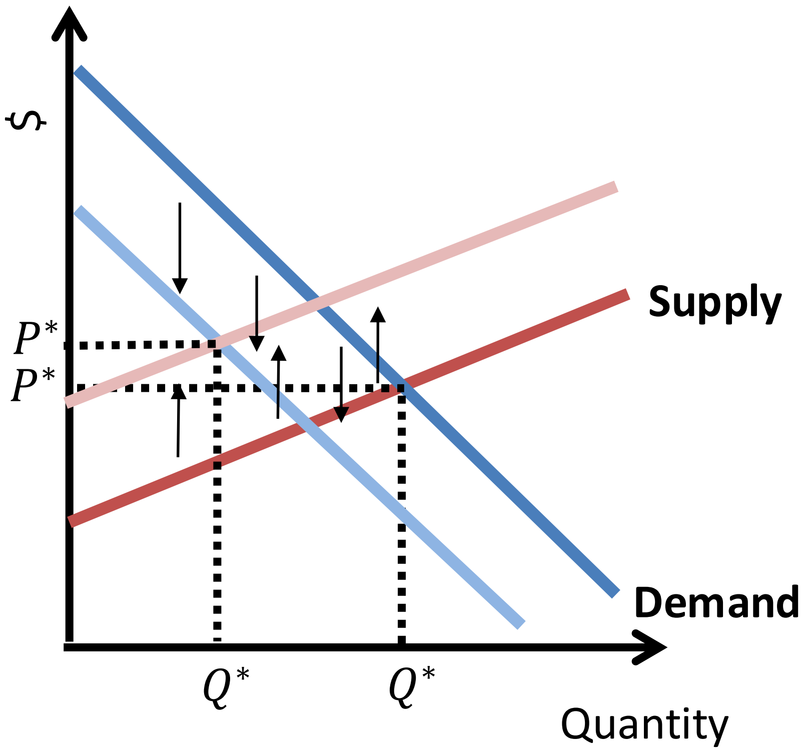
3. Did we converge towards the equilibrium?

… to be continued.

Demand and Supply Shifters

Demand Shifter

- When drawing a demand curve, everything is held constant except price and quantity.
- Many other factors can affect consumers’ reservation prices (Recall: TSA example).
- Same goes for supply.



Examples for “other factors”

- Demand
  - Prices of substitutes/complements
  - Population
  - Income
  - Product design
  - Advertising
- Supply
  - Input prices (more on this later)
  - Technology (more on this later)
  - Regulations (tax, license…)

Important to distinguish between:

- Movements **along** the curve (changes in price)
- **Shifts of the entire** curve (changes in other factors

Welfare Analyses

Motivation

We want to be able to think about how changes in a market affect consumers and producers.

- Are consumers helped or harmed from a shift in supply?
- How might consumers be affected by taxes/interventions?
- Can we compare how individuals fare under different policies?

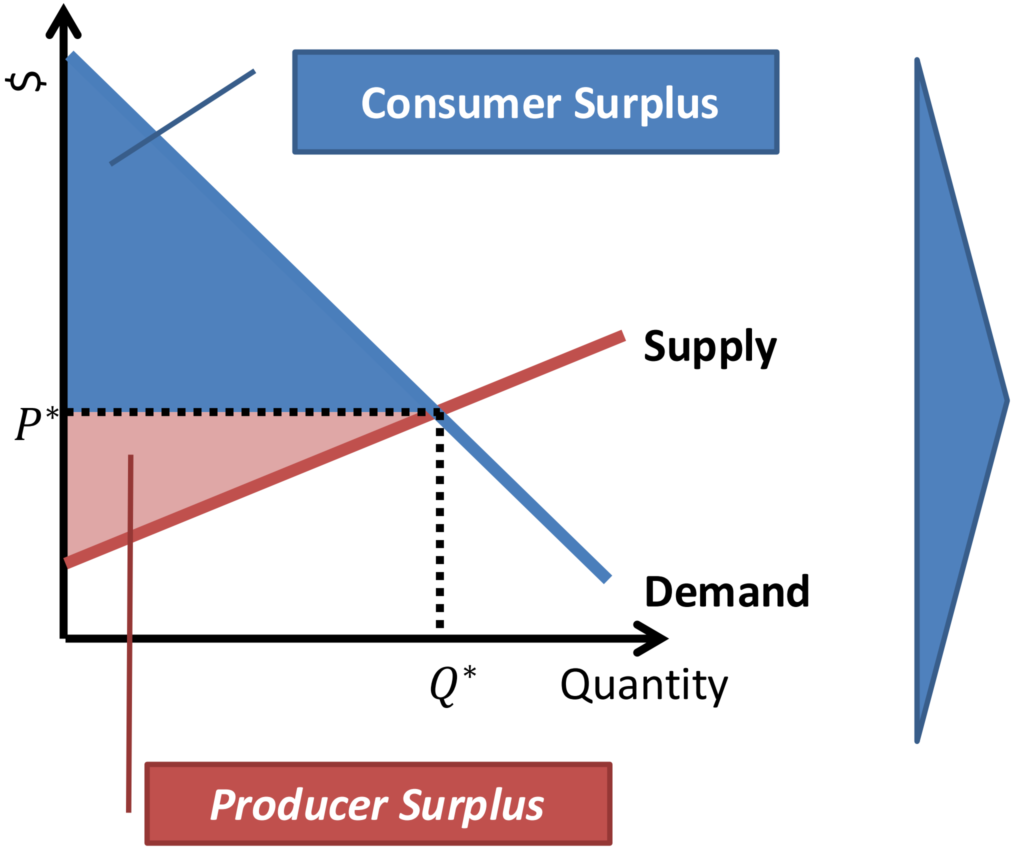
When analyzing a change in policy, we evaluate how the policy would change *Consumer Surplus* and *Producer Surplus*.

If *Consumer Surplus* increases from a policy, then that policy benefits consumers.
If *Producer Surplus* increases from a policy, then that policy benefits producers.

Consumer and Producer Welfare

Market Equilibrium

The price and quantity at which demand and supply intersect are the **market equilibrium**.
The market makes both consumers and suppliers better off, because some consumers pay less than their reservation price while some suppliers earn more than their cost.



Consumer Surplus:

- Sum of difference between reservation prices and the market price.

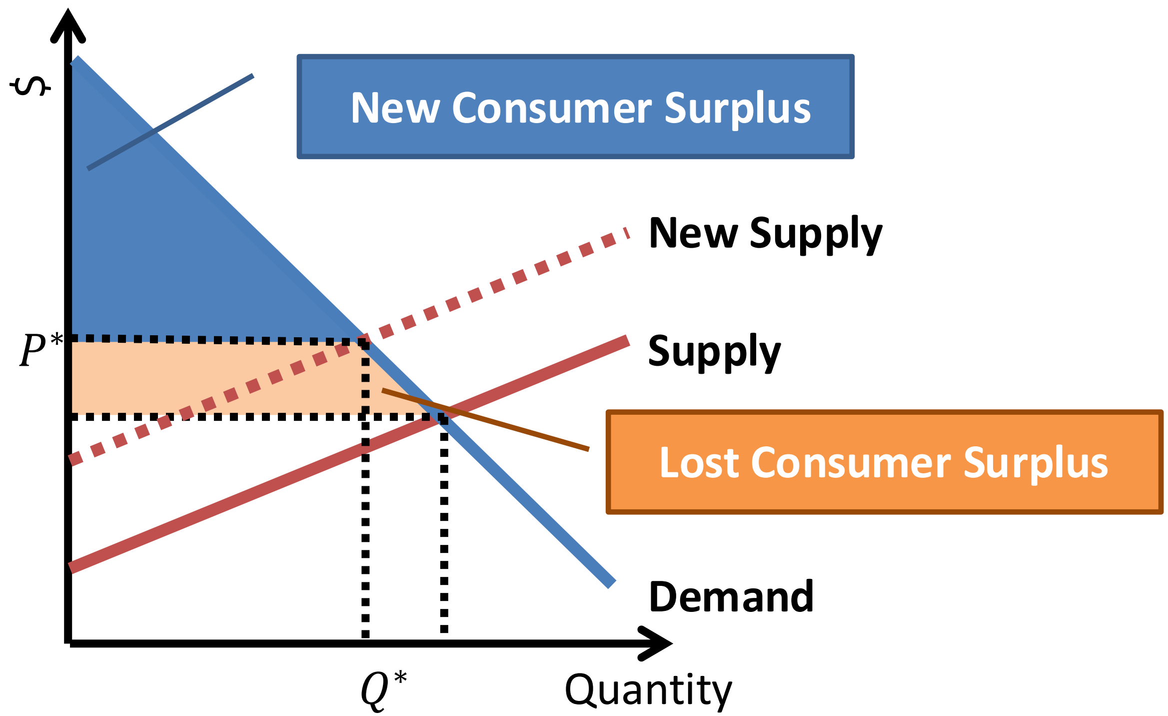
We also have Producer Surplus:

- Sum of difference between the market price and supply reservation prices.
- **Question: is this the same as profit?**

Using Consumer and Producer Surplus

Shifts in Market Equilibrium

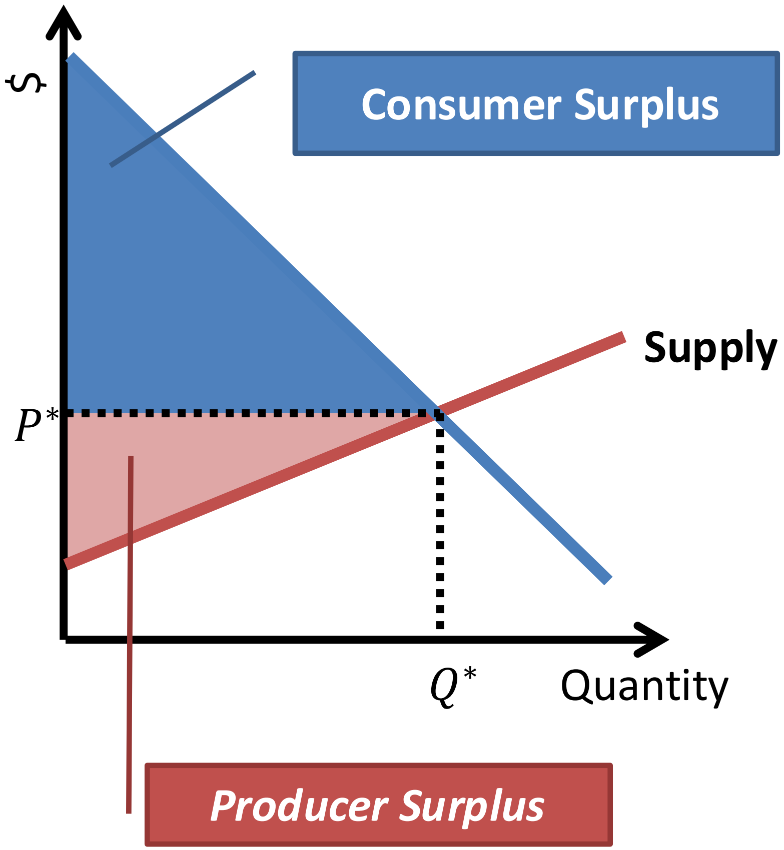
Suppose something changes in the market: input prices go up, leading to an upward shift of the supply curve:



We can use the concepts of consumer and producer surplus to evaluate how better or worse off each side of the market is after some change.

Why is the "equilibrium" point so great?

Let's take a look at a market that is in equilibrium: suppose price is P^*, and quantity produced/sold is Q^*.



Let's examine this outcome.
It there another price or quantity level that could increase both producer and consumer surplus?

The answer is no: the market has produced an "**efficient**" outcome. All of the possible gains from trade have been realized.

There is no buyer left willing to pay more than a seller's cost. There is no seller left willing to supply at a buyer's reservation price.

This is typically referred as the **first welfare theorem**, or "**the invisible hand**"

MobLab Discussion

Discussion points:

1. How did you do? Did you have a strategy?

2. Was the information provided (transactions) useful?

3. Did we converge towards the equilibrium?

4. What might prevent us from reaching the equilibrium?

In a competitive market, we should expect to end up in equilibrium, which is an **efficient** outcome. This means all potential gains from trade are being realized. However, any **frictions** in the market may lead to efficiency losses.

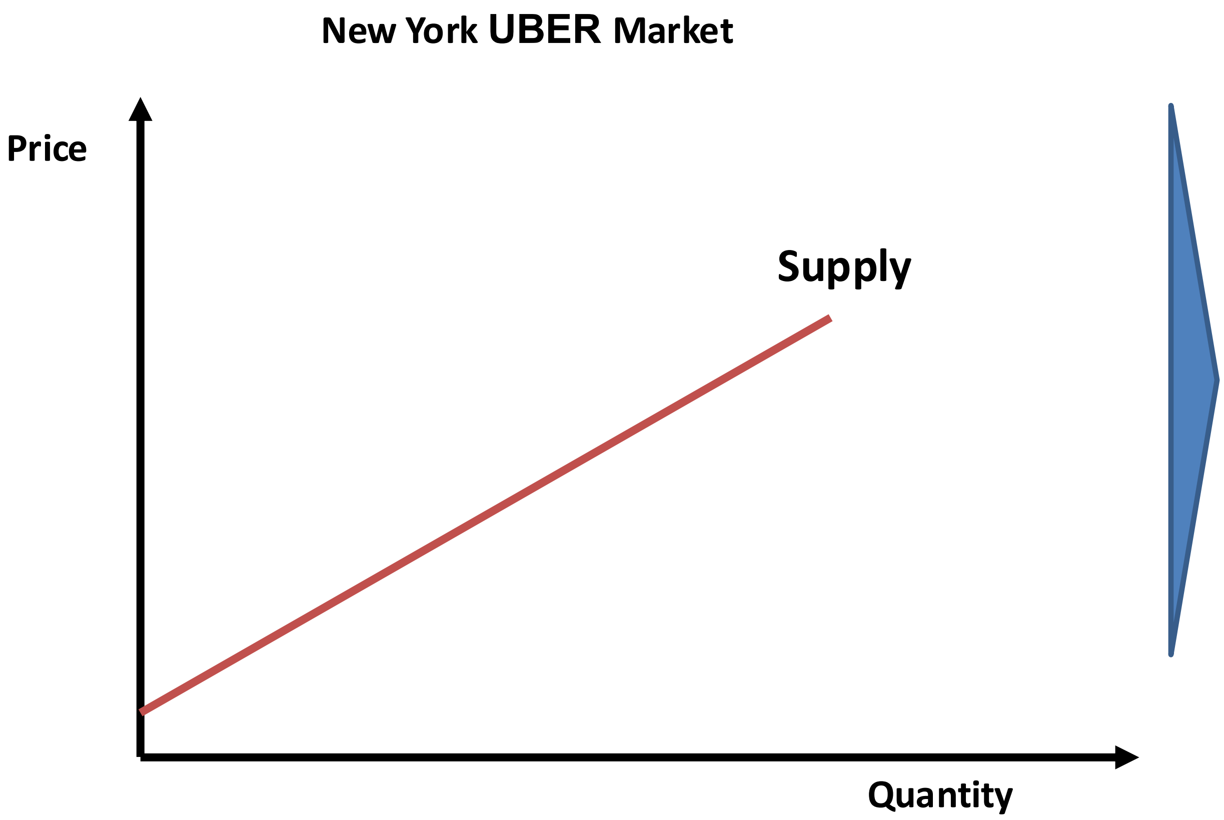
Today's Plan

Equilibrium and elasticity:

1. **Supply curves**
   - At what price is an individual or firm willing to supply a good or service?
2. **Equilibrium**
   - Equilibrium: when demand=supply
   - Consumer surplus and producer surplus
3. **Application to Uber**
   - **Uber surge pricing discussion**
   - **Finding the equilibrium price and quantity**

Article: Uber Dynamic Pricing

What does the UBER supply curve look like?

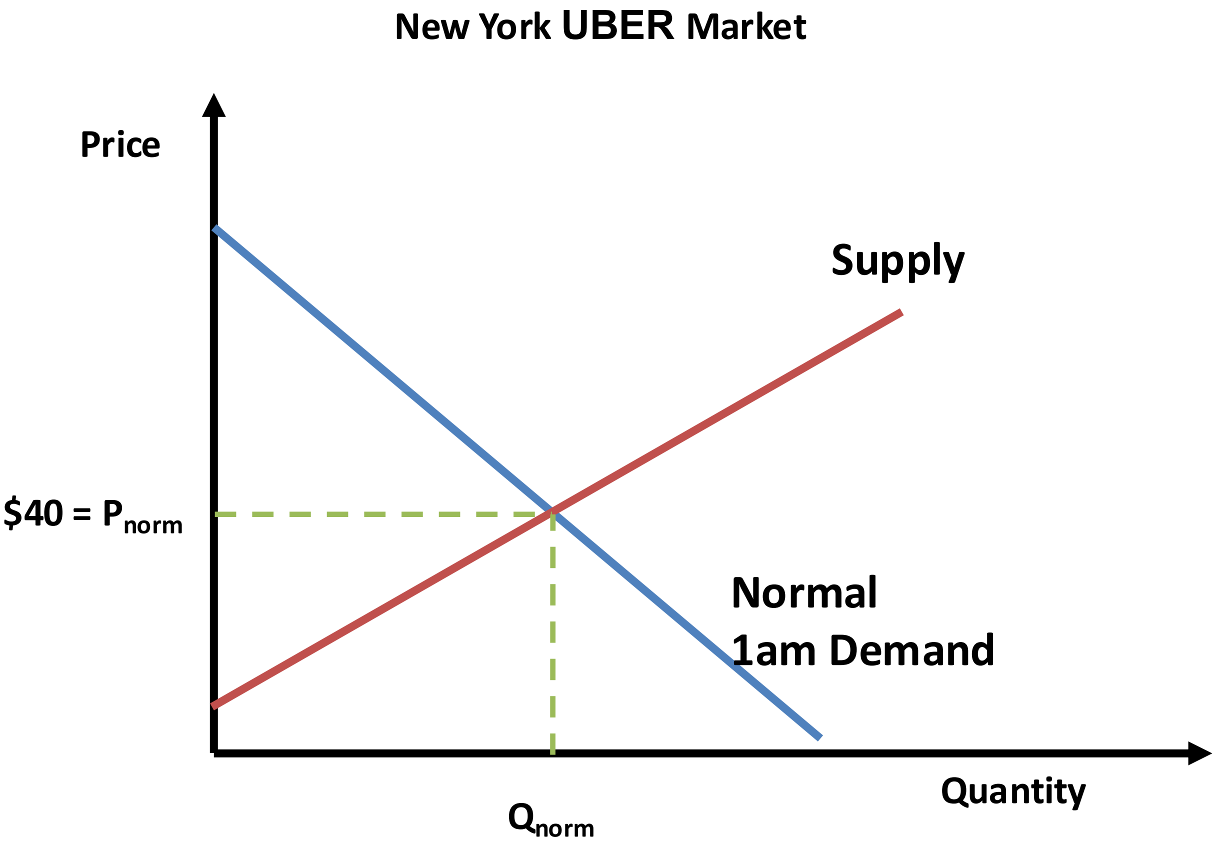


Note that supply is upward-sloping. Why?

- At a higher price, there are more drivers willing to set out on the roads.

Article: Uber Dynamic Pricing

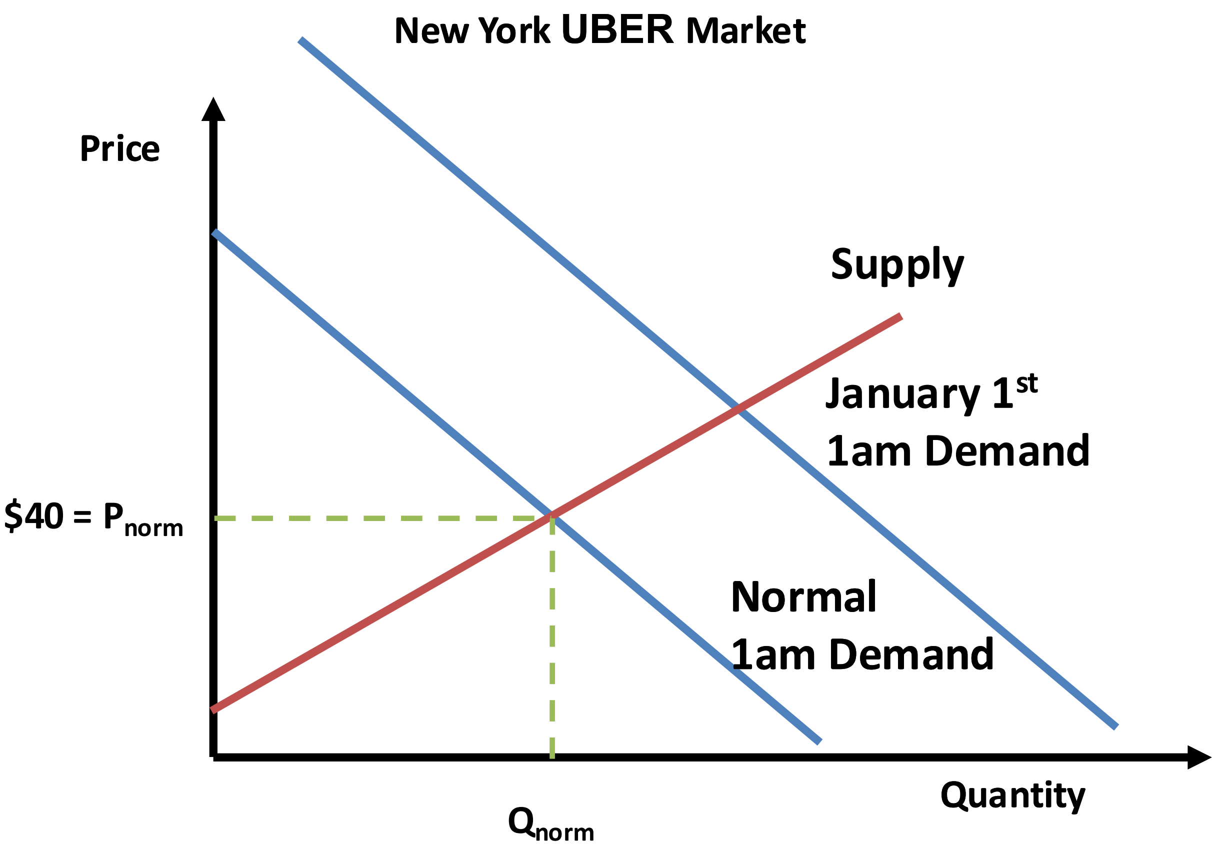
What does the typical 1am demand curve look like?



On a typical evening, we have an equilibrium price and quantity of, say, $40.

Article: Uber Dynamic Pricing

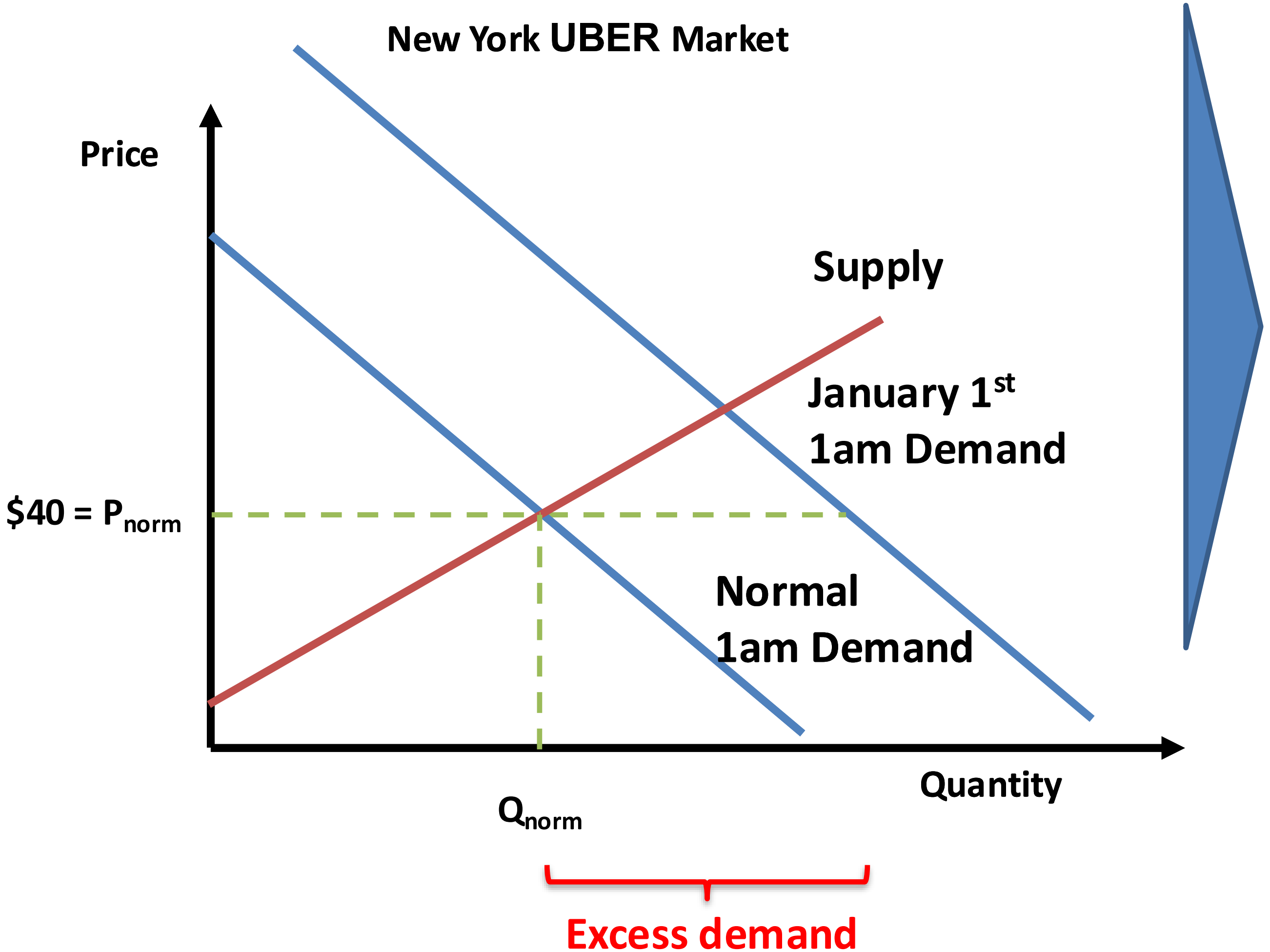
What does the New Year's Eve 1am demand curve look like?



On New Year's Eve, more people are willing to pay for a ride, and so the demand curve shifts outward.

Article: Uber Dynamic Pricing

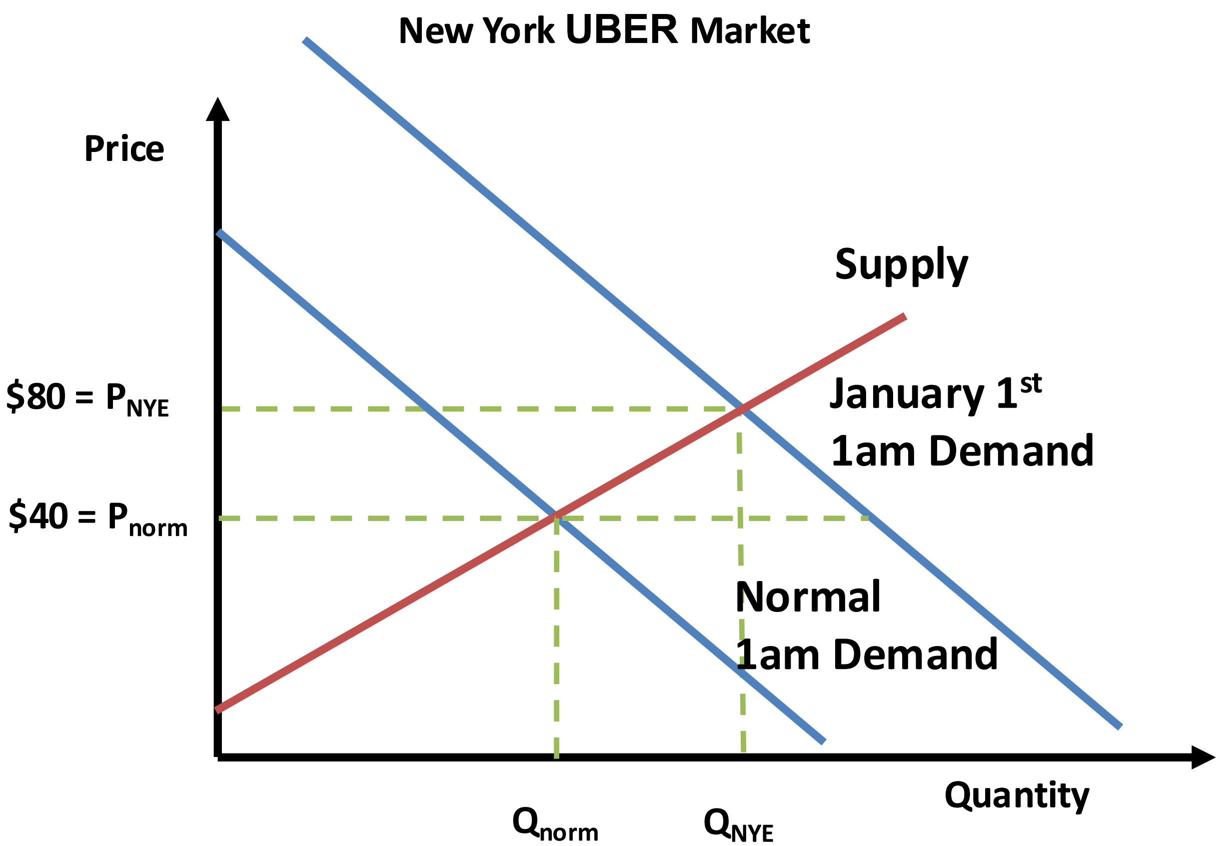
Suppose prices cannot adjust…



- Not everyone willing to pay the price will be able to get a car to pick them up.
- There is no way for those willing to pay more to “bid up” the price.
- “Rationing” of taxis during high demand is unlikely to match high reservation price customers to rides.

Article: Uber Dynamic Pricing

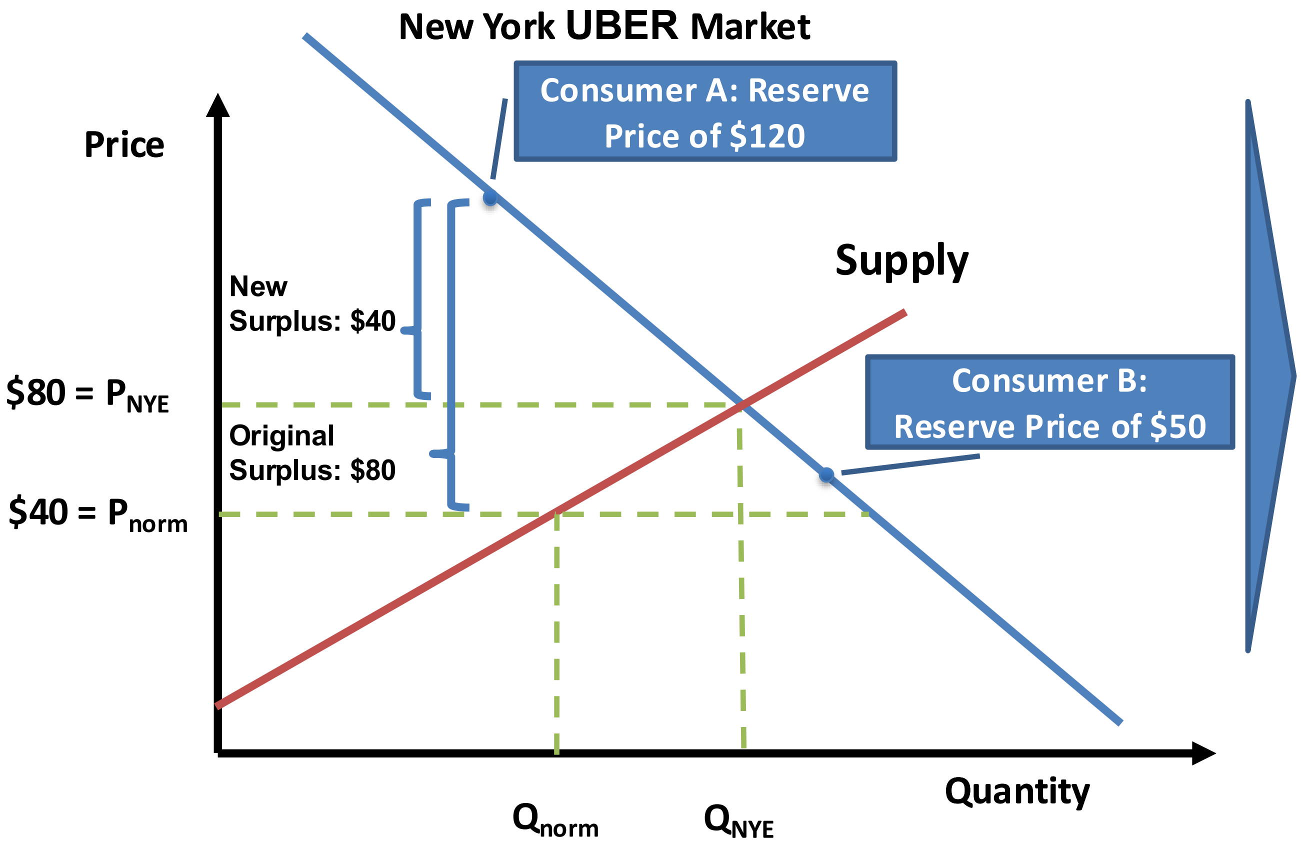
What does Uber's "Surge Pricing" do?



- They are trying to find a higher price that will "clear the market"
- At such a price, additional drivers will come out to help satisfy excess demand, and some consumers will no longer be willing to pay the price, further reducing excess demand.

Article: Uber Dynamic Pricing

What does Uber's "Surge Pricing" do?



- Consumer A might have not gotten a car before, but now will.
- However, lower surplus if he/she would have gotten a car before.
- Consumer B will not get a car now, but might have before: lower surplus.
- What if B didn't know the price had gone up (i.e. didn't notice surge pricing)?
- What happened to Producer Surplus?

Which version is "better" from an economics perspective?

Solving for a Market Equilibrium Mathematically

Suppose we have:
Market Demand

$$P_d = 180 - Q$$

Market Supply

$$P_s = 30 + 0.5Q$$

Equilibrium is the point (Q^*, P^*) where these two lines intersect.
Solve by setting $P_d = P_s$:

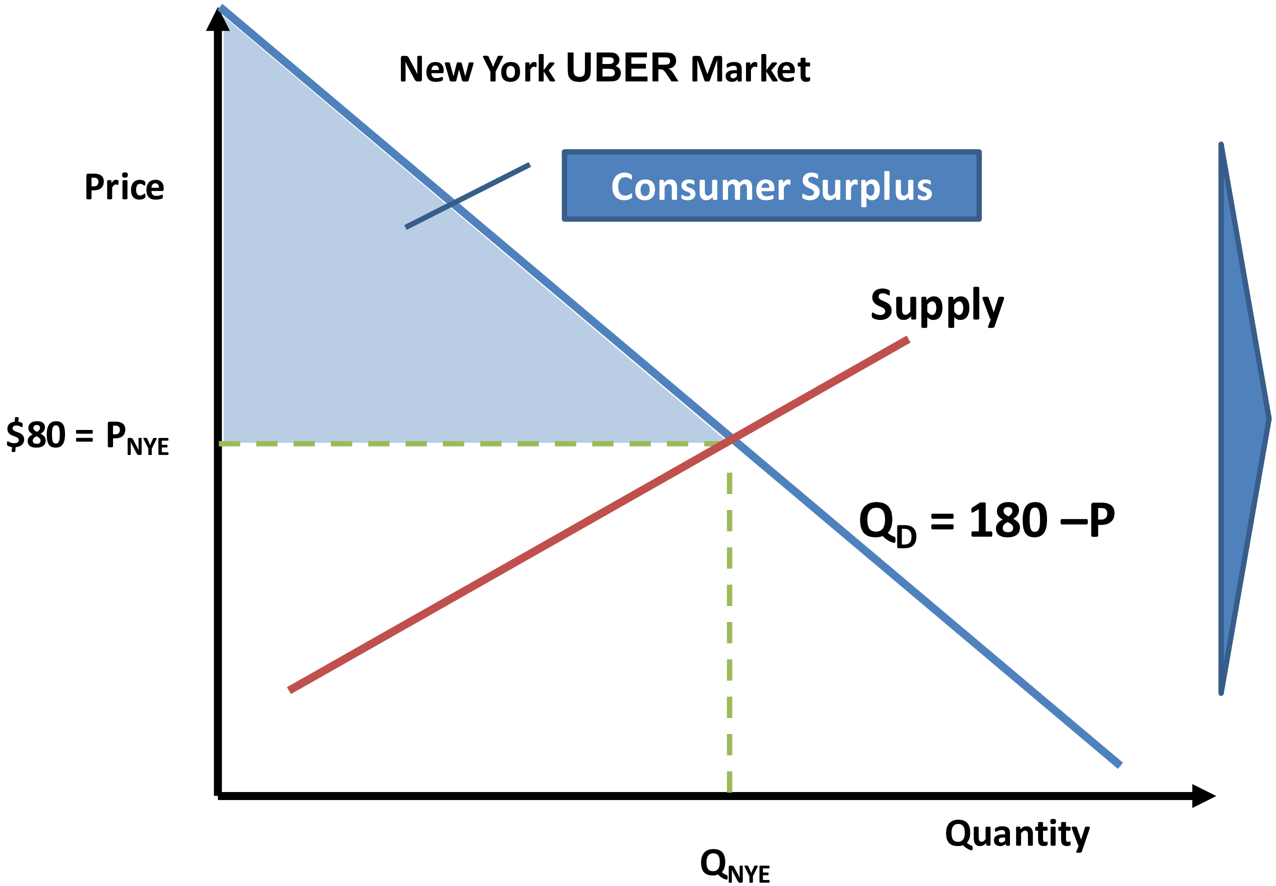
$$180 - Q = 30 + 0.5Q$$
$$150 = 1.5Q$$
$$Q^* = 100$$

Substitute back into (either) equation to get $P^* = 80$.
→The market equilibrium is a price of $80 and quantity of 100 units.

We can then evaluate a market equilibrium by computing measures such as consumer surplus and producer surplus.

Computing Consumer Surplus: Uber Example

What is the consumer surplus on New Year's Eve from Uber's service?
(demand is given by **Q_D = 180 –P**)



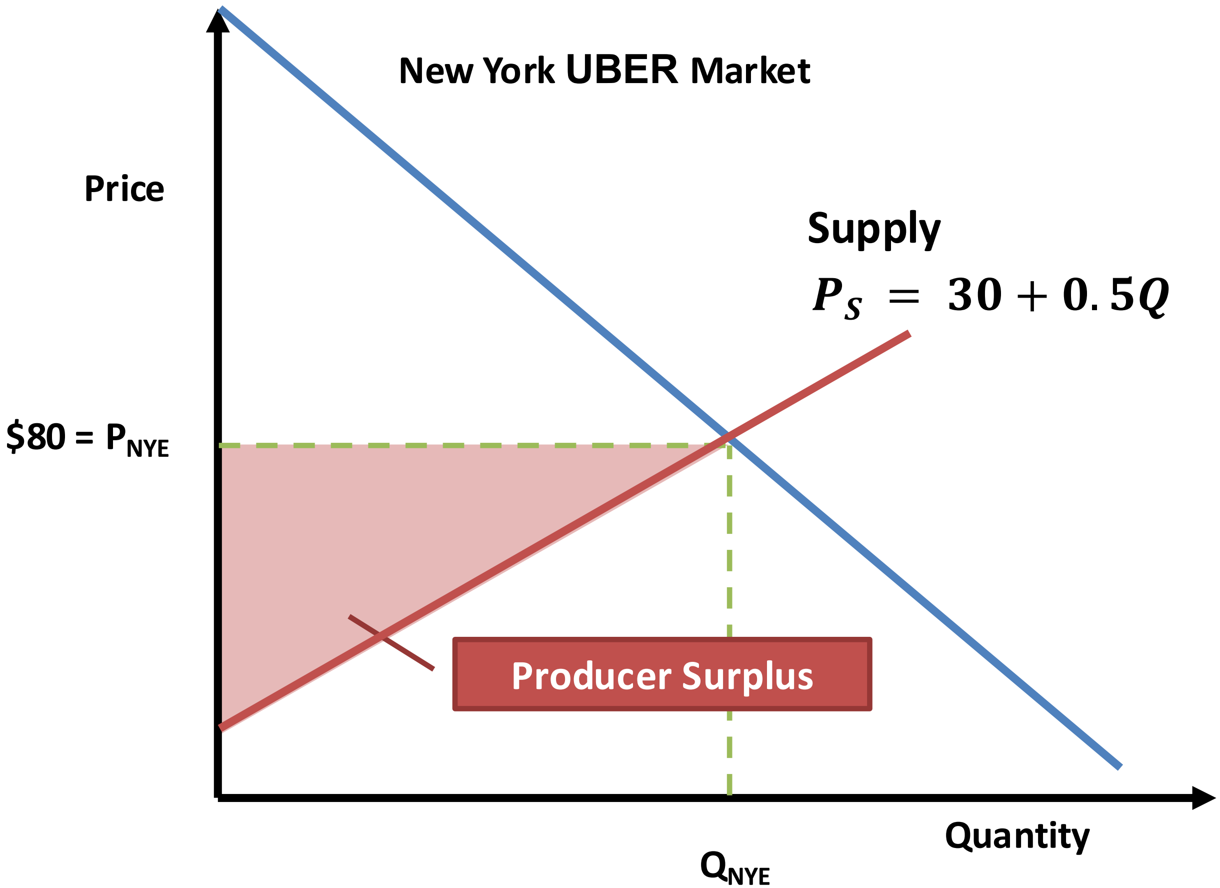
- With Linear demand, consumer surplus is the area of the right triangle.

$$CS = \frac{1}{2} \cdot base \cdot height$$
$$CS = \frac{1}{2} \cdot (100) \cdot (\$100)$$
$$CS = \$5{,}000$$

- With non-linear demand, the shape may not be a right triangle, but concept holds: it is the area between market price and demand.

Computing Producer Surplus: Uber Example

What is the **producer** surplus on New Year's Eve from Uber's service?
(supply is given by $\boldsymbol{P_S = 30 + 0.5Q}$)

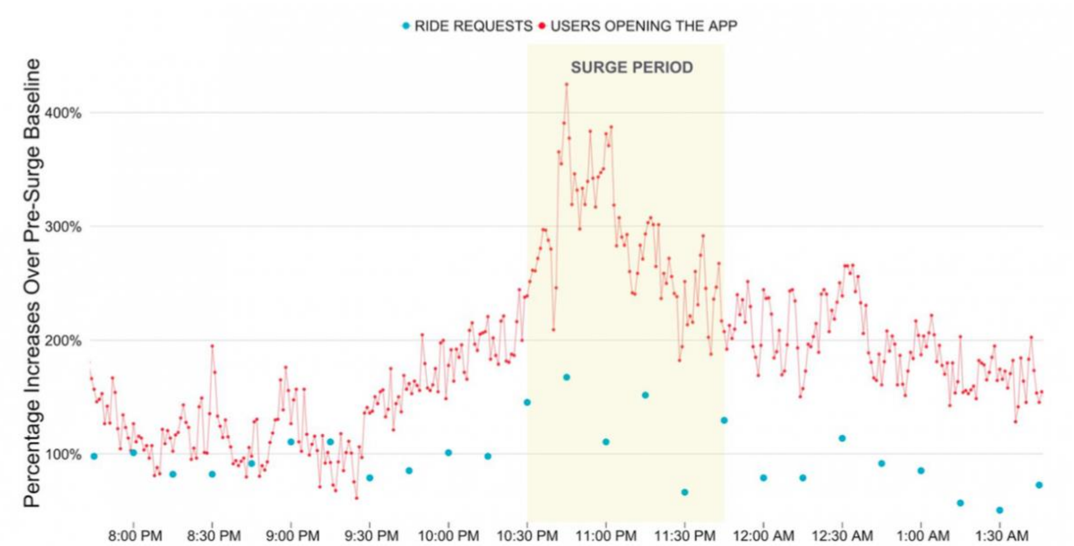


- With Linear Supply, producer surplus is the area of the right triangle.

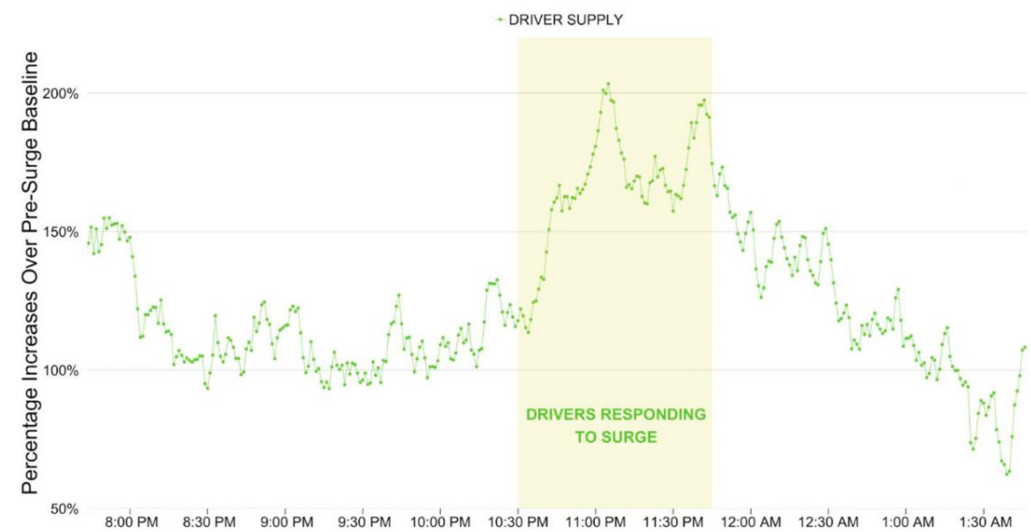
$$PS = \frac{1}{2} \cdot base \cdot height$$
$$PS = \frac{1}{2} \cdot (100) \cdot (\$50)$$
$$PS = \$2{,}500$$

- Non-linear supply?

Uber Article

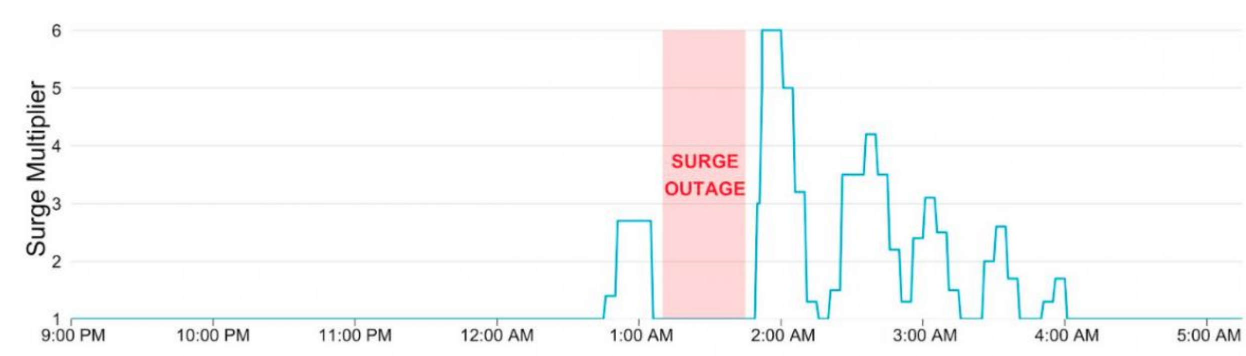


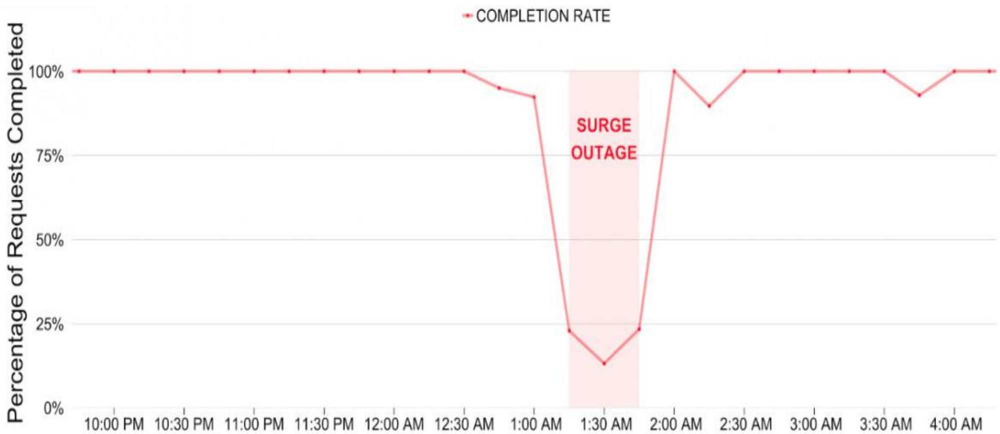
"Shift in Demand"



Upward-sloping supply of drivers

Uber Article





Price can no longer adjust to shifts in demand.

Not an equilibrium
- Rationing
- Not an efficient allocation.

Uber's "Surge Pricing" is a price-based approach to clear markets; it allocates rides to those that value them the most.

Why the Pushback?

Uber's surge pricing is far from popular.

- Common complaint is "price gouging", which arises in many situations.

Consumers don't like having to pay more than the "usual" price for a good.

- Paying more means less consumer surplus
- Actually very common for different people to pay different prices for the same good ("Price Discrimination", later in the course).

However, "price gouging" also likely increases supply.

- So, there are two sides to the story.
- Laws imposing price caps may actually **limit** supply to needed goods in emergencies.
- Most consumers don't think in equilibrium.

Good "price gouging": increases supply of essential items, allocates to those with higher need.
Bad "price gouging": transfers consumer surplus to producer (or intermediate) surplus, limits essential supplies to those with means to pay.

Price “gouging” to an economist:

When it increases supply and reallocates to those most in need…

The New York Times

DEALBOOK

Hurricane Price Gouging Is Despicable, Right? Not to Some Economists

Empty shelves at a supermarket in Florida. Supplies went quickly as Hurricane Irma approached.
Jason Henry for The New York Times

By Andrew Ross Sorkin

Sept. 11, 2017

When a devastating hurricane like Irma or Harvey arrives, stories about

Versus when it extracts consumer surplus from consumers to sellers (not even producers!):

The New York Times

Technology

He Has 17,700 Bottles of Hand Sanitizer and Nowhere to Sell Them

Jack Nicas

By Jack Nicas

1835 words
14 March 2020
05:00 AM
NYTimes.com Feed
NYTFEED
English

On March 1, the day after the first coronavirus death in the United States was announced, brothers Matt and Noah Colvin set out in a silver S.U.V. to pick up some hand sanitizer. Driving around Chattanooga, Tenn., they hit a Dollar Tree, then a Walmart, a Staples and a Home Depot. At each store, they cleaned out the shelves.

Over the next three days, Noah Colvin took a 1,300-mile road trip across Tennessee and into Kentucky, filling a U-Haul truck with thousands of bottles of hand sanitizer and thousands of packs of antibacterial wipes, mostly from “little hole-in-the-wall dollar stores in the backwoods,” his brother said. “The major metro areas were cleaned out.”

Summary and Direction

Today

Individual supply curves: comes from reservation prices, similar to demand but for selling instead of buying.
Market equilibrium: where supply and demand intersect.

Next Lectures

Elasticity:
- Own price elasticity
- Cross price elasticity
- Income elasticity